

(globals)

Debug Exp 8.cpp Exp9.cpp Exp10.cpp Exp11.cpp Exp12.cpp Exp13.cpp [*] Exp14.cpp Exp15.cpp Exp16.cpp Exp17.cpp

```
1  #include <stdio.h>
2  #define MAX_PROCESSES 5
3  #define MAX_RESOURCES 3
4  // Function Declaration
5  int is_safe(void); |
6  // Global Matrices
7  int available[MAX_RESOURCES] = {3, 3, 2};
8  int maximum[MAX_PROCESSES][MAX_RESOURCES] = {
9      {7, 5, 3},
10     {3, 2, 2},
11     {9, 0, 2},
12     {2, 2, 2},
13     {4, 3, 3}
14 };
15 int allocation[MAX_PROCESSES][MAX_RESOURCES] = {
16     {0, 1, 0},
17     {2, 0, 0},
18     {3, 0, 2},
19     {2, 1, 1},
20     {0, 0, 2}
21 };
22
23 int request_resources(int process_num, int request[]) {
24     // Check if request exceeds available resources or declared needs
25     for (int i = 0; i < MAX_RESOURCES; i++) {
```

Resources Compile Log Debug Find Results Close

Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\AADHITHYA\Exp17.exe
- Output Size: 130.4150390625 KiB
- Compilation Time: 1.00s

(globals)

Exp 8.cppExp 9.cppExp 10.cppExp 11.cppExp 12.cppExp 13.cpp[*] Exp 14.cppExp 15.cppExp 16.cppExp 17.cpp

25for (int i = 0; i < MAX_RESOURCES; i++) {
26 int need = maximum[process_num][i] - allocation[process_num][i];
27 if (request[i] > available[i] || request[i] > need) {
28 return 0; // Request cannot be granted
29 }
30}
31
32// Try allocating resources temporarily (Speculative allocation)
33for (int i = 0; i < MAX_RESOURCES; i++) {
34 available[i] -= request[i];
35 allocation[process_num][i] += request[i];
36 // FIXED: Removed 'maximum[i][i] -= request[i]'. Max resource needs do not change!
37}
38
39// Check if system is in safe state after speculative allocation
40if (is_safe()) {
41 return 1; // Request is officially granted
42} else {
43 // Roll back changes if it results in a unsafe state (Deadlock risk)
44 for (int i = 0; i < MAX_RESOURCES; i++) {
45 available[i] += request[i];
46 allocation[process_num][i] -= request[i];
47 }
48 return 0; // Request is denied
49}

Resources

Compile Log

Debug

Find Results

Close

tion

er paths

Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\AADHITHYA\Exp17.exe
- Output Size: 130.4150390625 KiB
- Compilation Time: 1.00s

(globals)

Exp 8.cpp Exp9.cpp Exp10.cpp Exp11.cpp Exp12.cpp Exp13.cpp [*] Exp14.cpp Exp15.cpp Exp16.cpp Exp17.cpp

```
49 | }
50 | }
51 |
52 | int is_safe(void) {
53 |     int work[MAX_RESOURCES];
54 |     int finish[MAX_PROCESSES] = {0};
55 |
56 |     // Initialize work array to current available resources
57 |     for (int i = 0; i < MAX_RESOURCES; i++) {
58 |         work[i] = available[i];
59 |     }
60 |
61 |     int count = 0;
62 |     while (count < MAX_PROCESSES) {
63 |         int found = 0;
64 |
65 |         for (int i = 0; i < MAX_PROCESSES; i++) {
66 |             if (finish[i] == 0) {
67 |                 int j;
68 |                 for (j = 0; j < MAX_RESOURCES; j++) {
69 |                     int need = maximum[i][j] - allocation[i][j];
70 |                     if (need > work[j]) {
71 |                         break;
72 |                     }
73 |                 }
```

resources CompileLog Debug Find Results Close

Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\AADHITHYA\Exp17.exe
- Output Size: 130.4150390625 KiB
- Compilation Time: 1.00s

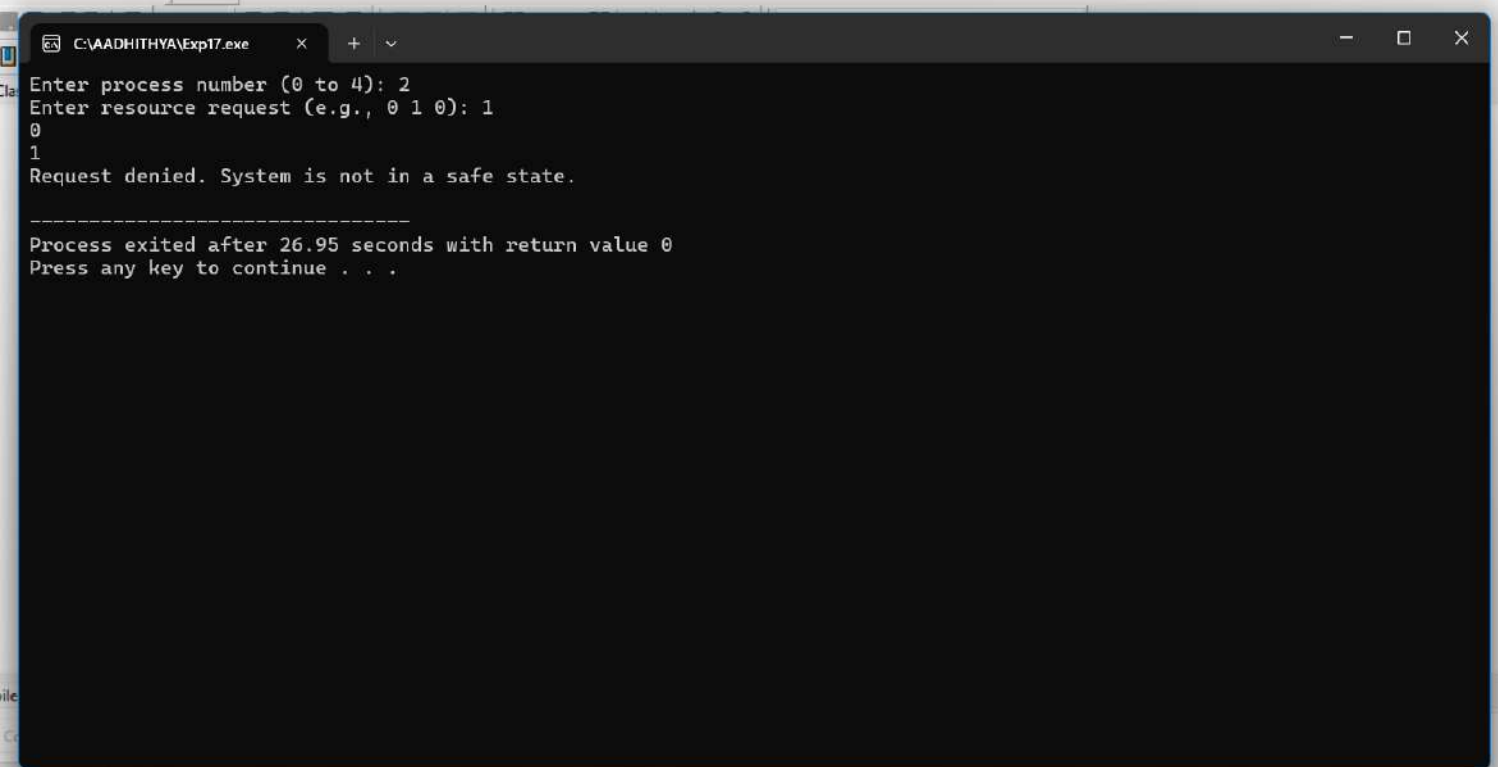
Exp 8.cpp Exp9.cpp Exp10.cpp Exp11.cpp Exp12.cpp Exp13.cpp [*] Exp14.cpp Exp15.cpp Exp16.cpp Exp17.cpp

```
67     int j;
68     for (j = 0; j < MAX_RESOURCES; j++) {
69         int need = maximum[i][j] - allocation[i][j];
70         if (need > work[j]) {
71             break;
72         }
73     }
74
75     // If all resources needed by process P_i are <= work available
76     if (j == MAX_RESOURCES) {
77         // Process can finish execution, release its allocated resources back to work
78         for (int k = 0; k < MAX_RESOURCES; k++) {
79             work[k] += allocation[i][k];
80         }
81         finish[i] = 1;
82         found = 1;
83         count++;
84     }
85 }
86
87
88 // If no progress was made in an entire loop, the system is in an unsafe state
89 if (found == 0) {
90     return 0;
91 }
```

ces  Compile Log  Debug  Find Results  Close

Compilation results...

```
-----
- Errors: 0
- Warnings: 0
- Output Filename: C:\AADHITHYA\Exp17.exe
- Output Size: 130.4150390625 KiB
- Compilation Time: 1.00s
```



```
C:\AADHITHYA\Exp17.exe
Enter process number (0 to 4): 2
Enter resource request (e.g., 0 1 0): 1
0
1
Request denied. System is not in a safe state.

-----
Process exited after 26.95 seconds with return value 0
Press any key to continue . . .
```

Compiler paths

- Warnings: 0
- Output Filename: C:\AADHITHYA\Exp17.exe
- Output Size: 130.4150390625 KiB
- Compilation Time: 1.00s